

Faculty of Technology and Engineering,
The Maharaja Sayajirao University of Baroda, Vadodara
51, Kalabhavan, Vadodara - 390001

Hack-A-Throne-25

PS 15:

UniNest: AI-Powered Campus Recruitment Portal

Problem Statement

The campus placement process today is fragmented, manual, and filled with inefficiencies—universities struggle with onboarding and verifying thousands of students, companies face delays in scheduling drives and screening talent, and students are left juggling incomplete resumes, missed opportunities, and lack of transparency.

UniNest aims to revolutionize this ecosystem by building a unified AI-powered recruitment portal where universities can seamlessly manage batches, filter students, and schedule drives, companies can smart-screen candidates and automate offers, and students can create verified digital profiles, build AI-enhanced resumes, and track every step of their placement journey.

The challenge is to design and deliver a platform that removes human error, ensures trust and fairness, and leverages modern tech to create a transparent, intelligent, and scalable placement experience—transforming campus recruitment into a streamlined, data-driven, and future-ready process.

1. Project Objective

UniNest is designed to **digitally transform university placement management** by connecting **Universities, Students, and Companies** on one unified platform.

The portal focuses on:

- **Universities** → Efficient student onboarding, drive scheduling, verification, and analytics.
- **Students** → Complete digital profile, AI-enhanced resumes, drive participation, and offer tracking.

- **Companies** → Simplified drive requests, smart student screening, and offer workflow. ●

Admins (UniNest Core Team) → Centralized monitoring, billing, and analytics.

The system removes manual errors, ensures transparency, and leverages AI to assist in decision-making.

2. Module-Wise Features

University Module

The **University** is the controlling authority for managing placement activities.

Student Onboarding & Management

- **Add Students in Multiple Ways**

- Bulk upload via **Excel/CSV** (predefined format).
- Manual entry using **student email ID** or **PRN No.**
- Share **credentials through email**

- **Verification Workflow**

- Pending student requests displayed in the dashboard.
- On first login, students must verify via **email OTP + phone OTP**.
- Once verified → student receives onboarding mail.

- **Automated Notifications**

- Students get an email: “You are added by University, please login and verify yourself.”

Department, Sub-Department & Class Management

- Create a **hierarchical structure**: Department → Sub-Department → Class (Keep fields). ●

Assign **Faculty Admins** with role-specific access.

- Maintain **class-wise student lists** for filtering drives.

Company & Drive Management

• **Company Profiles**

- Universities can create company profiles OR invite companies to create one. ◦

First login → company resets credentials & completes profile.

• **Placement Drives**

- Create drives in two ways:

1. **University-driven** → Directly create company profile + drive.

2. **Company-driven** → Company requests; University approves/rejects. ◦

Drives include: Title, JD, Salary, CGPA cutoff, Eligible Departments/Batches, etc. • **Student**

Filtering for Drives

- Multiple filters (Dept, Batch, CGPA, Skills).
- **AI Suggestion Engine** recommends best-fit students.
- Manual override option available. (Direct Entry by Admin).

• **Invitations & Access Levels**

- Invitations only initiated by University.
- Universities control whether companies can see invited/accepted students.

Resume Rules & Verification

- University sets **rules for resume creation**:

- Mandatory fields (CGPA, Projects, Certifications).

- Allowed templates (single/multiple).

- **Profile Verification**

- Fields can be marked as **Verified/Unverified**.
- If rejected → University leaves comment & reason.
- Students resubmit corrected info.
- **You can use the university API to fetch the student data to automate the verification process.**
- If university admin Locks student profile then student cannot update it. If a student wants to update then the student has to request the university.

- **Company View**

- Toggle:
 - ON → Companies can see unverified fields.
 - OFF → Only verified fields visible.
- “Verified by University” tag shown on approved documents.

Analytics & Insights

- Real-time dashboard for placement drives:
 - Students Invited / Accepted / Rejected.
 - Dept-wise participation.
 - CGPA/Gender distribution.
 - **AI-Driven Insights**
 - “20% of Civil Engineering students with >8 CGPA haven’t been invited.” ○
- Predictive stats → placement probability for each batch.

Student Module

The **Student** manages their digital identity, resumes, and placement journey.

Profile & Resume Management

- Full digital profile: Personal Info, Education, Skills, Projects, Achievements, Work Experience.
- Upload supporting **documents** (marksheets, certificates).
- Resume builder with **drag-and-drop sections**.
- Multiple **export templates** (PDF/DOC).
- Option to **export full profile** (A–Z student data).

AI Features for Students

- **ATS Resume-JD Matching**
 - Upload JD → system compares with resume.
 - Outputs **compatibility score** (0–100), keyword gaps, missing skills.
- **Resume Improvement Suggestions**
 - AI-generated tips for stronger resumes.
 - Suggests content rewrites for projects, skills, or achievements.

Placement Drive Participation

- Can see only drives invited by their University.
- Options: **Accept / Reject drive invite** (with reason).
 - Accepting adds them to “Drive Participants” viewable by companies (if university allows).

Offer Management

- Students view offers uploaded by companies.

- Actions:
 - Accept Offer.
 - Reject Offer (with feedback).
 - Counter-Offer (propose changes).
- History of negotiations tracked (Offer v1, v2, v3).

Verification & Transparency

- Students request verification for their profile.
- If rejected → see **reason comments** → correct & resubmit.
- Verified sections tagged clearly.
- New/updated fields need re-verification.

Company Module

The **Company** interacts with universities and students during the placement cycle.

Profile Management

- Complete profile with company details:
 - Name, Sector, Registration ID, Address, Logo, Website, Contact Person. ●

First login → **reset password** and confirm company details.

Drive Management

- **Create Drives**

- Define roles, upload JD, specify CGPA, skills, eligibility.
- Choose interview format (Online/Offline/Hybrid).

- **Request Drives**

- Request universities already listed in the portal.
- If university is not listed → invite via credentials.

Student Screening

- View students invited to a drive (as allowed by university).
- Download resumes (bulk/individual).

- **AI Smart Screening**

- Match scores for students vs JD.
- AI-generated summaries:

- Example: “Good fit for Backend Developer due to Java + SQL + Internship at TCS.”

- Mark students as **Shortlisted / Rejected** for tracking.

Offers Workflow

- Upload offers → students respond (accept/reject/counter).
- Offers tagged with **status labels**: Pending, Accepted, Rejected, Counter. ● AI

Offer Email Generator → auto-suggested formal email to student.

Admin Module (UniNest Core Team)

The **Admin** has centralized control across all universities, companies, and students.

University & Company Management

- Add new universities and companies.

- Approve/suspend tenants.
- Assign subscription plans and track billing.

Student & Drive Oversight

- View all students across universities.
- Global search/filter by CGPA, Skills, Department, Verification Status. •

Monitor all placement drives running across universities.

Analytics & AI Insights

- **Dashboard metrics:**
 - Total active universities, companies, students.
 - Drives conducted, students placed, offers accepted.
- **AI Placement Forecasting**
 - Predict which students/batches are most likely to be placed. •

Fraud Detection

- Detects duplicate resumes, fake docs, or inconsistent data.

3. Access Control

- **University → Student**
 - Invite, view, verify, and control visibility.
- **Company → Student**
 - View only university-approved data.
- **Student → Company**

- Limited company info (roles, drive details).

- **Admin → All**

- Full oversight with audit logs.

4. Key Highlights for Hackathon Participants •

Build **smart onboarding flows** (bulk upload + link invites).

- Create **intelligent filters** for student selection.
- Enhance **resume builder with AI improvements & ATS scoring**. •

Design **verification workflows** (approve/reject with comments). •

Develop **analytics dashboards** with predictive insights.

- Implement **student lifecycle tracking** (invite → drive → offer → acceptance). •

Add **AI-driven helpers**: Smart Screening, HelpBot, Offer Email Generator.

UniNest — Tech Stack & Delivery Spec (Hackathon)

5. Tech Stack (Preferrable)

- **DB**: Microsoft SQL Server (MSSQL)
- **Backend**: Node.js (LTS) + Express.js
 - ORM: Sequelize (raw SQL allowed)
 - Auth: JWT + OTP
- **Frontend**: React + Vite (fast) OR Next.js (default App Router)
- **API Testing**: Postman

- **Hosting:** Backend + Frontend both hosted
- **Alt Allowed:** NestJS + Prisma

6. Folder Structure

Backend

```
backend/  
  server.js  
  routes/  
  controllers/  
  services/  
  repositories/  
  models/  
  middlewares/  
  utils/  
  config/  
  .env  
  .example  
  package.json
```

Frontend (React)

```
frontend/  
  index.html  
  src/  
    main.jsx  
    App.jsx  
    components/  
    pages/  
    apis/  
    services/  
    hooks/  
    utils/  
  package.json
```

Frontend (Next.js)

- Use **default Next.js App Router structure**.

7. REST API Conventions

- Base: `/api/v1`

- Response:

```
{ "success": true, "data": {}, "message": "optional" }
```

- Core endpoints:

- `/students` (CRUD)
- `/companies` (CRUD)
- `/drives` (CRUD)
- `/universities` (CRUD)
- `/admin` (CRUD)

8. Postman Collection

- Deliver `UniNest.postman_collection.json`

- Folders: Students, Companies, Drives, University, Admin. •
Use variables:

◦ `{{base_url}}` = `http://localhost:8000/api/v1` ◦

`{{token}}` = JWT

9. GitHub Deliverable

- Public repo with `/backend` + `/frontend`

- Postman collection at root
- README.md with:
 - Project overview + flow
 - Tech stack
 - Setup steps (backend + frontend + Postman)
 - Folder structure explained
 - API flow (University → Company → Student)
 - Hosting guide

10. Checklist

- MSSQL + Node/Express + React/Next (or approved alt) • Follow backend layers (routes/controller/service/repo/models) • REST endpoints for students/companies/drives
- Postman collection included
- Both apps hosted & public
- README with clear flow + setup